Hyper-parameter Settings	
Behavioural encoder (collaborative filtering)	
Matrix factorisation	Truncated SVD
Latent components	64
Play-count transform	log(1 + x)
Acoustic encoder	
Audio features	13
Normalisation	MinMax + L2
Semantic encoder	
Representation	TF–IDF
n-gram range	1–2 (uni + bi)
Max vocabulary	3,000
Min document frequency	1
Fusion and scoring	
CF similarity	Dot product
Audio / semantic similarity	Cosine
Score normalisation	Min–max → [0, 1]
Simple-hybrid weights (b/a/s)	0.40 / 0.30 / 0.30
Proposed weights (b/a/s)	activity-adaptive
Evaluation and reproducibility	
Train/test split	Leave-one-out
Negatives per user	100
Ranking cut-offs (K)	5, 10, 20
Significance test	Paired t-test, Cohen's d
Random seed	42